

OS³ COMMAND CENTRE

Developer Handoff Document

Status snapshot — 2026-04-29

1. Project identity

OS³ Command Centre is the canonical operations surface for JB³Ai, owned and operated by Jono Blackburn (jono@jonoblackburn.com), based in Pretoria, Gauteng, South Africa.

It is one React application replacing a previously scattered set of HTML dashboards and a partial React app. It is the system of engagement; Supabase is the system of record.

Brand reminders (must)

- All product references use JB³Ai and OS³ with the superscript ³. No JB3Ai, no OS3.
- Cyan #2AF1FF is the accent, capped at ~15% pixel coverage.
- Premium Gold #D4AF37 is reserved for money/exposure callouts only.
- Orbitron for headlines and key metric numbers; Inter for body; JetBrains Mono for code/monospace.
- No gradients, no glow, no ornament for ornament's sake.

2. Architecture overview

A single Vite + React 19 + TypeScript SPA, deployed to Vercel, talking to one Supabase project for auth, data and RLS, with Anthropic Claude Sonnet 4.6 for AI briefings, Kestra for orchestration of ingest pipelines, and a small constellation of MCPs (apple-mcp, whatsapp-bridge, filesystem-mcp) for native-app and messaging integrations.

Layer	Technology	Notes
Frontend	React 19, Vite 8, TypeScript ~6, Tailwind 3.4	Path alias @/* -> ./src/*. Brand tokens in tailwind.config.js.
UI primitives	Radix (shadcn-style), lucide-react, cmdk, date-fns	Resizable component removed (incompatible with react-resizable-panels v4).
Auth/DB	Supabase JS 2.103, project uxeolplwhtyyefpmwktw (eu-west-1)	Magic-link login. RLS enforced on every new table.
Hosting	Vercel (team Jono's projects)	GitHub-Vercel webhook auto-deploys on push to main.
AI	Anthropic claude-sonnet-4-6 for briefings; Ollama (Gemma 3n / Qwen 3) locally for experimentation	Azure OpenAI is used elsewhere; not used inside Command Centre yet.
Workflows	Kestra (Docker, localhost:8080)	NOT n8n. Phase 3 in progress, currently blocked.

Native bridge	apple-mcp on iMac via Cloudflare Tunnel; WhatsApp Go bridge + Python MCP	Always-on infra dependency.
---------------	--	-----------------------------

3. Repositories and deployment

Concern	Where
GitHub repo (this project)	https://github.com/JB3Ai/jb3ai-command-centre — private
Local working tree	C:\Apps in Dev Visual Code Folder\Claude Working Folder\PROJECTS\jb3ai-command-centre
Sibling repo (separate, untouched)	C:\Apps in Dev Visual Code Folder\jb3ai-os3 -> https://github.com/JB3Ai/jb3ai-os3
Vercel team	team_Ad0clhIFpEafbbmJozJ4a7zz (slug jonos-projects-cdacc0ad)
Vercel project	prj_ee5aX5JwEw0RO0RpE21swpYf88Z
Production URL	https://jb3ai-command-centre-jonos-projects-cdacc0ad.vercel.app
Branch alias (always points at main)	https://jb3ai-command-centre-git-main-jonos-projects-cdacc0ad.vercel.app
Latest production deploy (2026-04-29)	dpl_C4pxrbGmbWmunE47tZqXJDUZRhGX (commit a1f984b)
Custom domain	Originally planned: command.jb3ai.com. DROPPED on 2026-04-28 — we ship on the auto Vercel domain.

vercel.json (root of repo)

Pinning the framework, the SPA rewrite, security headers, and immutable cache for /assets/*. The SPA rewrite is critical — React Router uses BrowserRouter, so any deep-link to /config or /braveheart 404s without the rewrite. Re-read this file when modifying build config.

```
{
  "$schema": "https://openapi.vercel.sh/vercel.json",
  "framework": "vite",
  "buildCommand": "npm run build",
  "outputDirectory": "dist",
  "installCommand": "npm install",
  "rewrites": [
    { "source": "/((?!assets/).*)", "destination": "/index.html" }
  ],
  "headers": [
    { "source": "/(.*)", "headers": [
      { "key": "X-Frame-Options", "value": "DENY" },
      { "key": "X-Content-Type-Options", "value": "nosniff" },
      { "key": "Referrer-Policy", "value": "strict-origin-when-cross-origin" },
      { "key": "Permissions-Policy", "value": "camera=(), microphone=(), geolocation=()" }
    ] },
    { "source": "/assets/(.*)", "headers": [
      { "key": "Cache-Control", "value": "public, max-age=31536000, immutable" }
    ] }
  ]
}
```

4. Environment variables

Three Vite-prefixed variables are required for both local dev and prod. The Supabase anon key is safe to publish (RLS gates everything); the file src/lib/supabase.ts also keeps a hardcoded fallback for the URL and anon key so the app boots even without .env.local.

Variable	Purpose	Example
VITE_SUPABASE_URL	Supabase project endpoint	https://uxeolplwhtyyefpmwktw.supabase.co
VITE_SUPABASE_ANON_KEY	Public anon JWT (RLS-gated)	eyJ... (paste from .env.local)
VITE_AUTH_ALLOWED_EMAILS	Comma-separated whitelist for magic-link	jono@jonoblackburn.com

Set the same three in Vercel -> Settings -> Environment Variables, applied to Production + Preview + Development. Vercel changes do not retro-apply — a redeploy is needed for new values to land.

Supabase Auth URL configuration

- Site URL: https://jb3ai-command-centre-jonos-projects-cdacc0ad.vercel.app
- Additional Redirect URLs: prod URL/auth/callback, branch-alias URL/auth/callback, http://localhost:5173/auth/callback
- Auth -> Providers -> Email -> Allow signups OFF, invite-only — only addresses on the allowed-emails list can ever sign in.

5. Database schema

Project: uxeolplwhtyyefpmwktw (eu-west-1). Schema lives in supabase/migrations/ and is applied via the Supabase SQL editor (not yet automated via supabase-cli). All migrations are additive — they extend existing hub_* tables rather than dropping or rewriting columns.

Existing tables (extended in 001, kept)

hub_tasks, hub_calendar, hub_emails, hub_deploys, hub_braveheart, hub_subscriptions, hub_subscriptions_meta, hub_briefings, hub_productivity_sessions, hub_sync_status, hub_automation_log.

Notably, hub_braveheart was extended (not forked) with: account_ref, priority (Critical/High/Medium/Low), response_status (red/amber/green/unknown), on_credit_record, imed_status, action_status, attorney_name, attorney_meta (jsonb), last_correspondence_at, updated_at.

hub_sync_status was extended with: display_name, category (mcp/api/script/scheduler/tunnel/db/ingest/other), enabled, endpoint, latency_ms.

New tables (created in 001)

Table	Powers panel	Key columns
hub_daily_intentions	HOME / Productivity	for_date (unique), intention
hub_daily_reviews	Productivity	for_date (unique), win_1, win_2, win_3, lesson, mood

hub_reading_queue	Productivity	title, url, source, read_at, archived, tags[]
hub_links	Links	label, url, category, pinned, sort_order, notes
hub_notes_dump	Notes	body, source, source_ref, pinned, archived, tags[]
hub_quick_capture	HOME quick-capture (-> ClickUp Inbox)	body, pushed_to_clickup, clickup_task_id
hub_marketing_leads	Marketing	name, email, company, source, stage, exposure_zar
hub_news_items	News	title, url, source (rss:* or gmail:ai-news), summary
hub_media_items	Media kanban	title, kind, status (idea->done), prompt, asset_url
hub_monthly_chronicles	Chronicle	for_month (unique), title, highlights, body_md
hub_whatsapp_messages	WhatsApp	jid, direction, body, sent_at
hub_bankzero_transactions	BankZero (Phase 4)	txn_date, description, amount_zar, balance_zar

Migration files

File	What it does
001_initial_os3_schema.sql	Adds columns to hub_braveheart and hub_sync_status; creates 12 new tables; enables RLS + auth-only policies on the new tables only.
002_seed_creditors.sql	UPSERTs 24 creditor rows into hub_braveheart from creditors-dashboard.html.
003_seed_integrations.sql	UPSERTs 21 integrations into hub_sync_status (4 MCPs, 8 APIs, 1 tunnel, 2 schedulers, 1 db, 5 ingest pipelines). Idempotent.

Sensitive-data rule: creditor names, account numbers, claim amounts, BankZero balances, and attorney details NEVER appear in code, commits, memory files, or cross-conversation context. They live only in Supabase, RLS-protected.

6. Routes and panels

14 routes in the left-nav, all wrapped in <ProtectedRoute><Layout/>. Plus /productivity (separate panel, not in nav), /login (public), /auth/callback (public).

Route	Status	Phase	Source tables
/home	Built (Phase 2)	2	hub_calendar, hub_emails, hub_tasks, hub_daily_intentions, hub_briefings

/braveheart	Built (Phase 2)	2	hub_braveheart (24 seeded rows)
/bankzero	Placeholder	4	hub_bankzero_transactions
/whatsapp	Placeholder	3	hub_whatsapp_messages
/subscriptions	Built (Phase 2)	2	hub_subscriptions, hub_subscriptions_meta
/ecosystem	Built (Phase 2)	2	hub_sync_status
/projects	Built (Phase 2)	2	hub_deploys
/chronicle	Placeholder	3	hub_monthly_chronicles
/media	Placeholder	3	hub_media_items
/marketing	Placeholder	3	hub_marketing_leads
/news	Placeholder	3	hub_news_items
/notes	Built (Phase 2)	2	hub_notes_dump
/links	Built (Phase 2)	2	hub_links
/config	Built (Phase 1)	1	hub_sync_status
/productivity	Built (legacy)	—	hub_productivity_sessions, hub_briefings, hub_tasks

Panel pattern (used by all Phase 2 + CONFIG panels)

- Brand-aligned card layout (Tailwind utilities, not inline styles).
- Top: icon halo + Orbitron title + description; right side has manual Refresh and last-sync pill.
- Summary strip with key counts; gold callouts for money/exposure only.
- Inline error banner (rose) when a Supabase query errors.
- Manual refresh only — no auto-polling — except ProductivityPanel which still polls every 5 minutes.
- Reads via `supabase.from('hub_*').select(...)`. Most panels also support write-back where it makes sense (Notes, Links, HOME daily intention, CONFIG enable toggle).
- Empty/loading/error states for every list.

Productivity (separate)

`src/components/ProductivityPanel.tsx` is older code that uses inline styles and a `COLORS` object instead of Tailwind utility classes. It works and is shipped. Do not refactor it just for consistency — wait until a real change is needed there.

7. File map (src/)

```
src/  App.tsx                                <- LEGACY   main.tsx
Route tree, AuthProvider wrap, BrowserRouter  index.css
Tailwind directives + Inter/Orbitron/JetBrains Mono Google Font links  components/
PlaceholderPage.tsx                  Used by every Phase 3/4 route still on hold
ProductivityPanel.tsx                Older inline-style component, mounted at /productivity
ProtectedRoute.tsx                  Redirects to /login if no Supabase session      layout/
Layout.tsx                          Sidebar + Topbar + StatusBar wrapping <Outlet/>
Sidebar.tsx                          Reads NAV_ITEMS from src/lib/nav-config.ts (14 entries)
```

```

Topbar.tsx                                User menu, logout                                StatusBar.tsx
Bottom strip      ui/                      shadcn components; resizable.tsx was
REMOVED on 2026-04-28  lib/                supabase.ts                                Single client
instance + Hub* row types  auth-context.tsx                                AuthProvider + useAuth
hook      nav-config.ts                                NAV_ITEMS array — source of truth for sidebar
utils.ts                                cn() helper for Tailwind class composition  pages/
home/                                Phase 2 panel (rebuilt 2026-04-29)  braveheart/
Phase 2 panel      subscriptions/          Phase 2 panel      projects/
Phase 2 panel      links/                  Phase 2 panel      notes/
Phase 2 panel      ecosystem/              Phase 2 panel      config/
Phase 1 panel      bankzero/              Phase 4 placeholder  whatsapp/
Phase 3 placeholder  chronicle/            Phase 3 placeholder  media/
Phase 3 placeholder  marketing/            Phase 3 placeholder  news/
Phase 3 placeholder  login/                Public, magic-link entry
auth-callback/          Public, magic-link landing supabase/migrations/
001_initial_os3_schema.sql  002_seed_creditors.sql  003_seed_integrations.sql
vercel.json              Build + SPA rewrite + security headers
tsconfig.json              "ignoreDeprecations": "6.0" + baseUrl + paths
tsconfig.app.json          Same fix; React app build tsconfig.node.json
vite.config.ts only tailwind.config.js      Brand tokens
(matte/graphite/steel/edge/cyan/gold/ink) + animations

```

8. Build pipeline — lessons baked into the repo

These are gotchas that have already been solved; the fixes are committed. They are listed here so you don't accidentally undo them or get caught by them in a fresh checkout.

TypeScript 6 baseUrl deprecation

TypeScript 6 still warns when baseUrl is set, and Vercel's CI escalates that warning to an error, failing the build. Fixed by adding "ignoreDeprecations": "6.0" to both tsconfig.json and tsconfig.app.json. Don't remove baseUrl — the @/* path alias depends on it.

shadcn resizable.tsx vs react-resizable-panels v4

shadcn's default resizable component imports ResizablePrimitive.PanelGroup/PanelResizeHandle, which were removed in react-resizable-panels v4. Since nothing in the app actually uses Resizable, the file was deleted on 2026-04-28. If you ever need a resizable layout: `npx shadcn@latest add resizable` will drop a v4-compatible version.

Vercel "Require Verified Commits" gate

Jono's Vercel team has Require Verified Commits enabled at the team level. Every webhook-triggered deploy with an unsigned commit is INSTANTLY canceled with empty build logs — a confusing failure mode. Fixed by overriding to Disabled at the project level (Settings -> Git). Future hardening: set up SSH-signed commits and re-enable.

Vercel Redeploy reuses the original SHA

Clicking Redeploy on an old failed deployment rebuilds that SAME commit — not the latest main. To rebuild against latest main: push a fresh commit (an empty commit is fine: `git commit --allow-empty -m "trigger rebuild"`), or in the dashboard pick the row whose SHA matches the latest main commit.

9. Phase status (as of 2026-04-29)

Phase 1 — Foundation: DONE

- Supabase auth wired (magic link), 3 env vars in prod.
- Schema applied (migrations 001-003) including 24 creditor rows + 21 integration rows.
- Vercel project + GitHub-Vercel webhook live.
- Left-nav shell with 14 placeholder routes.
- CONFIG / Integrations panel built with heartbeat + per-row toggle persisted to `hub_sync_status`.

Phase 2 — Already-have-data panels: DONE

- HOME (calendar/emails/tasks/intention/latest briefing).
- BRAVEHEART (24 creditor rows with priority, exposure in gold, response status, credit-record flag, expandable detail).
- Subscriptions (status lights, monthly + annual rollup in gold, grouped by category).
- Projects (Vercel deploys grouped by project, state dots, build durations).
- Links (CRUD + pin + category groups).
- Notes (quick capture + tags + pin/archive filter chips).
- Ecosystem (tile grid view of `hub_sync_status`, complementing CONFIG).

All seven shipped in commit `a1f984b` on 2026-04-29 — build green, deployed to production.

Phase 3 — Kestra ingest pipelines: IN PROGRESS, BLOCKED

5 flows planned: News RSS -> `hub_news_items`, Monthly Chronicle -> `hub_monthly_chronicles`, Media kanban refresh -> `hub_media_items`, Marketing leads -> `hub_marketing_leads`, WhatsApp logs -> `hub_whatsapp_messages`.

Kestra is running locally on Docker (localhost:8080), pulling Ollama models from the Mac host via `host.docker.internal:11434`.

Current blocker: Kestra's `io.kestra.plugin.ai.completion.ChatCompletion` uses type: `USER` (not role: `USER`) in its messages array. The first test flow (`gemma_test` in `jono.lab` namespace) still has role: `USER` and fails on every execution with `IllegalArgumentException: Unrecognized field "role"`. The fix is one-word per message; saving in the Kestra Source tab is what was missing on the last attempts.

Secondary issue noted: the test flow uses `modelName: gemma4:e4b`. There is no Gemma 4 — the intended model is `gemma3n:e4b` (multimodal Gemma 3n with effective ~4B params). Verify against ollama list on the Mac before assuming.

Tertiary: the prompt uses `<|think|>` which is a Qwen 3 control token, not Gemma. On Gemma it is treated as literal text. Use a plain "first reason step-by-step then conclude" prompt, or swap the provider to Qwen 3.

Phase 4 — Hard integrations: PENDING

- BankZero: no public API — manual CSV upload only. UI not built.
- cPanel UAPI: works but locked behind login — needs an authenticated bridge.
- Meta FB/IG: requires app review with business verification (months of work) — deferred.

10. Locked decisions (in chronological order)

Date	Decision
2026-04-25	One React app, one Supabase project, single Vercel deploy. No self-host.

2026-04-25	Anthropic claude-sonnet-4-6 for briefings (NOT Azure OpenAI inside Command Centre).
2026-04-25	Kestra for workflows. n8n explicitly excluded.
2026-04-25	Productivity gadgets locked: Next-up, Quick capture, Daily intention, Daily review, Reading queue. Streak tracker dropped.
2026-04-25	ClickUp is output-only on free tier. NOT the primary system of record.
2026-04-26	Magic-link email login from Phase 1.
2026-04-26	Existing hub_* tables KEPT and EXTENDED additively, never forked.
2026-04-26	Creditor data lives in extended hub_braveheart, NOT a new hub_creditors table.
2026-04-26	CONFIG dashboard reuses hub_sync_status, NOT a new hub_integrations_status.
2026-04-28	GitHub repo is JB3Ai/jb3ai-command-centre (private) — NOT jb3ai-os3 which is a separate older repo.
2026-04-28	Custom domain command.jb3ai.com DROPPED. Using Vercel auto-domain.
2026-04-28	Vercel "Require Verified Commits" set to Disabled at project level.
2026-04-29	7 Phase 2 panels shipped to production.

11. Open issues / loose ends

1. Kestra Phase 3 blocker: change role: USER -> type: USER on the gemma_test flow, save (the Save action in the Kestra Source tab is separate from Execute), then re-execute. Also fix model name to gemma3n:e4b.
2. Diagnostic row cleanup pending: delete from public.hub_sync_status where source = 'test:diagnostic'; — left over from the 003 seed debugging session.
3. JS bundle size warning: 591 kB raw / 166 kB gzipped. Default Vite warning at 500 kB. Fix when convenient via route-level dynamic import() to code-split each panel.
4. Verified-commit gate currently disabled. Long-term hardening: set up SSH-signed commits on the Windows dev machine and re-enable the gate.
5. Legacy src/App.tsx still in repo, orphaned by the HOME rebuild. Safe to delete after smoke-test.
6. ProductivityPanel.tsx uses older inline-style pattern; other panels use Tailwind utility classes. Inconsistency, not a bug.
7. Phase 3 panels (news, chronicle, media, marketing, whatsapp) still PlaceholderPage — build them after the corresponding Kestra flow lands data in their tables.
8. BankZero panel still PlaceholderPage. Phase 4.

12. Working with the repo

Local development


```
cd "C:\Apps in Dev Visual Code Folder\Claude Working Folder\PROJECTS\jb3ai-command-
centre" npm install cp .env.example .env.local # then fill in the 3 vars npm
run dev # http://localhost:5173
```

Build verification

```
npm run build # tsc -b && vite build # Expected output
(approx): # 2158 modules transformed. # dist/index.html 0.92 kB #
dist/assets/index-*.css 53.71 kB # dist/assets/index-*.js
591.26 kB # (warning about >500 kB is known; not blocking)
```

Deploy

git push origin main — the GitHub-Vercel webhook auto-builds and promotes on success. To trigger a fresh build without code changes: git commit --allow-empty -m "trigger rebuild" && git push.

Watching a Vercel build

The connected Vercel MCP exposes list_deployments, get_deployment, get_deployment_build_logs, get_runtime_logs, web_fetch_vercel_url. There is no env-var setter or domain creator — those still need a quick dashboard click.

13. This session's build history (2026-04-29 chronological)

9. Started by writing 003_seed_integrations.sql (21 integration UPSERTs, atomic, idempotent, preserves live poller state).
10. Built CONFIG / Integrations panel: header, summary, sectioned heartbeat list, per-row enable toggle persisted, manual refresh.
11. Layout fix to CONFIG: stacked rows instead of cramped single-line meta strip; header collapses below md breakpoint.
12. Kicked off Vercel deploy: wrote vercel.json + .env.example + hardened .gitignore.
13. git init / first commit (a45febe) — push to GitHub. Vercel imported, framework auto-detected as Vite.
14. First deploy ERRORED on TS5101 baseUrl deprecation. Fix: ignoreDeprecations: 6.0 + remove unused shadcn resizable.tsx -> commit 9233d02.
15. 9233d02 webhook deploy CANCELED with empty logs. Diagnosed as Require Verified Commits team policy. Disabled at project level.
16. Empty commit c8be187 -> first GREEN deploy. Production live.
17. Set 3 env vars in Vercel -> redeploy ed87b5c with vars baked in. Updated Supabase Auth Site URL + Redirect URLs. Magic-link confirmed working in prod.
18. Dropped the command.jb3ai.com custom domain plan in favour of the auto vercel.app URL.
19. Phase 2: built 7 panels (HOME, BRAVEHEART, Subscriptions, Projects, Links, Notes, Ecosystem). +2160 lines of TSX. Local build green. Commit a1f984b. Vercel auto-rebuild green.
20. Started Phase 3 Kestra. Wrote test flow gemma_test in jono.lab namespace. First execution failed: role: USER not recognised. Diagnosed the role/type mismatch and the gemma4 -> gemma3n typo. Re-execute attempted but Save step in Kestra UI was missed — same error returned. Currently blocked here.

14. Immediate next action: unblock Kestra

In the Kestra UI for flow gemma_test (namespace jono.lab):

21. Go to the Source tab (not Topology, not Executions).
22. Replace the entire Source content with the YAML below.
23. Click Save first — it is a separate action from Execute. Without Save, the editor changes are not applied.
24. Click Execute. Expected result: a successful run with the model's response logged by the log_response task.

```
id: gemma_test namespace: jono.lab tasks:
  - id: chat_with_gemma type: io.kestra.plugin.ai.completion.ChatCompletion provider: io.kestra.plugin.ai.provider.Ollama modelName: gemma3n:e4b baseUrl: http://host.docker.internal:11434 messages:
    - type: USER content: "First reason step-by-step about the trade-offs, then conclude: why is a 10th Gen i9 desktop chip a better server host than an 11th Gen i7 mobile chip for 24/7 AI orchestration?"
  - id: log_response type: io.kestra.plugin.core.log.Log message: "{{ outputs.chat_with_gemma.choices[0].message.content }}"
```

Verify the model is pulled on the Mac (ollama list). If gemma3n:e4b is not present: ollama pull gemma3n:e4b. Verify Ollama is reachable: curl http://localhost:11434/api/tags from the Mac shell.

15. Suggested next steps for the developer

25. Smoke-test all 7 Phase 2 panels in production. Confirm RLS + magic-link give the right view and the data shapes match what the panels expect.
26. Unblock Kestra: apply the role -> type fix to gemma_test, save, execute — confirm a green run.
27. Build Kestra ingest flow #1 — News RSS poller -> hub_news_items — with the role/type lesson baked in.
28. Wire the /news panel to read from hub_news_items.
29. Repeat for Monthly Chronicle, Media kanban, Marketing, WhatsApp.
30. Phase 4: BankZero CSV upload UI -> hub_bankzero_transactions; cPanel auth bridge if/when needed.
31. Polish pass: route-level code-splitting; retire src/App.tsx; SSH commit signing + re-enable Verified Commits gate; clean up the test:diagnostic row in hub_sync_status.

Contact

Owner: Jono Blackburn — jono@jonoblackburn.com. WhatsApp 071 969 1848 (no unscheduled calls; bookings only).